

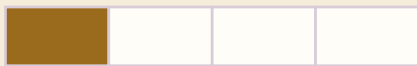
Understanding fractions

Fractions describe equal parts of a whole or a quantity. In Year 3, pupils explore unit and non-unit fractions, tenths, equivalence, comparison, and simple addition and subtraction.

Unit and non-unit fractions

A unit fraction has a numerator of 1. A non-unit fraction has a numerator greater than 1. The denominator tells how many equal parts make the whole.

UNIT FRACTION

 $\frac{1}{4}$ 

1 equal part selected

NON-UNIT FRACTION

 $\frac{3}{4}$ 

3 equal parts selected

ESSENTIAL RULE Fraction parts must be equal in size.

Fractions in everyday life

 $\frac{3}{8}$ **PIZZA**

3 of 8 equal slices

 $\frac{1}{2}$ **JUICE**

half a cup

 $\frac{1}{4}$ **SHARING**

one equal share each

 $\frac{3}{4}$ **RACE**

three-quarters completed

Tenths connect fractions to measure and money

MEASUREMENT

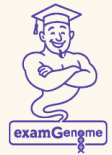
$\frac{1}{10}$ of 1 metre = 10 cm

MONEY

$\frac{1}{10}$ of £1 = 10p

CORE IDEA

A fraction names equal parts: the denominator names the parts; the numerator counts them.



Reasoning with fractions

Equivalent fractions - different names, same value

Equivalent fractions describe the same amount. One-half and two-quarters cover the same proportion even though they are written differently.

ONE-HALF



$\frac{1}{2}$

=

TWO-QUARTERS



$\frac{2}{4}$

Comparing fractions

For unit fractions, more equal parts make each part smaller. With the same denominator, compare the numerators because every part has the same size.

UNIT FRACTIONS

$$\frac{1}{3} > \frac{1}{4}$$

thirds are larger parts

SAME DENOMINATOR

$$\frac{3}{8} > \frac{1}{8}$$

three parts are more than one

Adding and subtracting like fractions

When denominators are the same, the parts are equal in size. Add or subtract the numerators and keep the denominator unchanged.

PACKED LUNCHES

$$\frac{3}{7} + \frac{1}{7} = \frac{4}{7}$$

add equal sevenths

PARTS REMAINING

$$\frac{6}{8} - \frac{2}{8} = \frac{4}{8}$$

subtract equal eighths

By the end of this topic, pupils should be able to:

- count up and down in tenths
- recognise fractions as numbers on a number line
- compare unit fractions and like fractions
- recognise unit and non-unit fractions
- identify simple equivalent fractions
- add and subtract fractions with the same denominator

LOOKING AHEAD

Secure fractions prepare pupils for decimals, percentages, ratio, proportion and algebra.